

TEPS School Distinction - Preparation (Final Revision)

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```
## Load packages
library(tidyverse)
library(haven)
```

1. Student Survey Data

```
## Read student data
wls <- read_dta("data_raw/w1_j_s_lv7.0.dta") # wave 1 (2001), n = 20055
#sapply(wls, function(y) sum(is.na(y))) %>% data.frame(na_count = .) %>% filter(na_count != 0)

## Gender (male=1)
wls$male <- case_match(wls$wls502, 1 ~ 1, 2 ~ 0, 99 ~ NA)
#wls %>% count(male)

## Siblings
for (i in 1:4) {
  wls[[c('edbro', 'ygbro', 'edsis', 'ygsis')[i]]] <- case_match(
    wls[[c('wls203', 'wls204', 'wls205', 'wls206')[i]]],
    1 ~ 0, c(2, 3, 4, 5) ~ 1, c(97, 99) ~ -5)
  #wls %>% count(edbro)

  wls$sib <- wls$edbro + wls$ygbro + wls$edsis + wls$ygsis # all NA = -20
  wls$onlychild <- case_match(wls$sib, 0 ~ 1, -20 ~ NA, .default = 0)
  #wls %>% count(onlychild)

  wls$edsib <- wls$edbro + wls$edsis - wls$onlychild # those with no elder siblings and not the only child
  wls$edstchild <- case_match(wls$edsib, 0 ~ 1, NA ~ NA, .default = 0)
  #wls %>% count(edstchild)

  wls$ygsib <- wls$ygbro + wls$ygsis - wls$onlychild # those with no younger siblings and not the only child
  wls$ygstchild <- case_match(wls$ygsib, 0 ~ 1, NA ~ NA, .default = 0)
  #wls %>% select(edbro, edsis, ygbro, ygsis, onlychild, edstchild, ygstchild)

## Cognitive Ability
wls$cogability <- wls$w1all3p
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## Geographical Area
wls <- wls %>% mutate(geoarea = case_match(wladmarea,
  c(11,31,21) ~ 1, # BBG
  c(33,22,34,35) ~ 2, # TZZM
  c(23,36,37,38) ~ 3, # ZZZT
  c(39,24,40,25,41) ~ 4, # YJJNN
  c(12,42,43) ~ 5, # GGP
  c(32,45,44,46) ~ 6)) # YHDP

## Select variables
wls <- wls %>% select(
  stud_id, wlsch_id, w1priv, w1urban3, w1scarea, wladmarea, geoarea, # student & school info.
  male, onlychild, edstchild, ygstchild, cogability) # student background & outcome
```

2. Family Survey Data

```
## Read family data
wlp <- read_dta("data_raw/wl_j_p_lv7.0.dta") # n = 19779

## Filter data
wlp <- wlp %>% filter(wlp101 == 1) # responded by one of the parents, n = 19053
#wlp %>% count(wlp101)

## Marriage
wlp$marriage <- case_match(wlp$wlp103, c(1,2) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(marriage)

## Ethnicity (1TW, 2MainLand, 3Indi&Other)
wlp$faethn <- case_match(wlp$w1faethn, c(1,2) ~ 1, 3 ~ 2, c(4,5) ~ 3, c(97,99) ~ 9)
wlp$moethn <- case_match(wlp$w1moethn, c(1,2) ~ 1, 3 ~ 2, c(4,5) ~ 3, c(97,99) ~ 9)
#xtabs(~faethn + moethn, wlp) %>% addmargins(FUN = sum)

wlp <- wlp %>%
  mutate(pethn = case_when(
    faethn == 1 & moethn == 1 ~ 1, # both TW
    faethn == 1 & moethn >= 3 ~ 2, # 1 TW + 1 Non-Main
    faethn >= 3 & moethn == 1 ~ 2,
    faethn == 2 & moethn == 2 ~ 3, # both MainLand
    faethn == 2 | moethn == 2 ~ 4, # 1 MainLand
    faethn == 3 & moethn == 3 ~ 5, # Other (both Indi/Other)
    .default = NA))
#wlp %>% count(pethn)

wlp <- wlp %>% mutate(faethn = case_match(faethn, 9 ~ NA, .default = faethn)) %>%
  mutate(moethn = case_match(moethn, 9 ~ NA, .default = moethn))
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## Education
wlp$faedu <- case_match(wlp$w1faedu, c(6,97,99) ~ 0, .default = wlp$w1faedu)
wlp$moedu <- case_match(wlp$w1moedu, c(6,97,99) ~ 0, .default = wlp$w1moedu)
#xtabs(~ faedu + moedu, wlp) %>% addmargins(FUN = sum)

wlp <- wlp %>%
  mutate(pedu = case_when(
    faedu >= moedu ~ faedu,
    faedu < moedu ~ moedu)) %>%
  mutate(pedu = case_match(pedu, 0 ~ NA, .default = pedu))
#wlp %>% count(pedu)

wlp$pedu2 <- case_match(wlp$pedu, c(1,2,3) ~ 0, c(4,5) ~ 1, NA ~ NA)
#wlp %>% count(pedu2)

wlp <- wlp %>% mutate(faedu = case_match(faedu, 0 ~ NA, .default = faedu)) %>%
  mutate(moedu = case_match(moedu, 0 ~ NA, .default = moedu))

## Income (<2, 2-5, 5-10, 10-15, >15)
wlp$income <- case_match(wlp$w1p515, 6 ~ 5, c(97,99) ~ NA, .default = wlp$w1p515)
#wlp %>% count(income)

wlp$income2 <- case_match(wlp$income, c(1,2,3) ~ 0, c(4,5) ~ 1, NA ~ NA)
#wlp %>% count(income2)

## Parent SES (Education x Income)
wlp$pses <- case_when(
  wlp$pedu2 == 0 & wlp$income2 == 0 ~ 1,
  wlp$pedu2 == 0 & wlp$income2 == 1 ~ 2,
  wlp$pedu2 == 1 & wlp$income2 == 0 ~ 3,
  wlp$pedu2 == 1 & wlp$income2 == 1 ~ 4)

## Occupation (1Labor, 2Service, 3Public, 4Expert, 5Other)
wlp$faocc <- case_match(wlp$w1faocc, 7 ~ 1, 6 ~ 2, c(1,2) ~ 3, c(3,4,5) ~ 4, c(8,9) ~ 0, c(97,99) ~ -1)
wlp$moocc <- case_match(wlp$w1moocc, 7 ~ 1, 6 ~ 2, c(1,2) ~ 3, c(3,4,5) ~ 4, c(8,9) ~ 0, c(97,99) ~ -1)
#xtabs(~ faocc + moocc, wlp) %>% addmargins(FUN = sum)

wlp <- wlp %>%
  mutate(pocc = case_when(
    faocc == 4 | moocc == 4 ~ 4, # 4: Exp
    faocc == 3 | moocc == 3 ~ 3,
    faocc == 2 | moocc == 2 ~ 2,
    faocc == 1 | moocc == 1 ~ 1,
    faocc == 0 | moocc == 0 ~ 5, # 5: Other
    faocc == -1 & moocc == -1 ~ NA))
#wlp %>% count(pocc)

wlp <- wlp %>% mutate(faocc = case_when(
  faocc == -1 & moocc != -1 ~ 5,
  faocc == -1 & moocc == -1 ~ NA,

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faocc == 0 ~ 5,
.default = faocc))
wlp <- wlp %>% mutate(moocc = case_when(
  moocc == -1 & faocc <= 5 ~ 5,
  moocc == -1 & is.na(faocc) ~ NA,
  moocc == 0 ~ 5,
  .default = moocc))

## Employer
wlp$faboss <- case_match(wlp$wifaself, c(3,4) ~ 1, c(97,99) ~ 9, .default = 0)
wlp$moboss <- case_match(wlp$wimoself, c(3,4) ~ 1, c(97,99) ~ 9, .default = 0)
#xtabs(~faboss + moboss, wlp)

wlp <- wlp %>%
  mutate(pboss = case_when(
    faboss == 1 | moboss == 1 ~ 1,
    faboss == 9 & moboss == 9 ~ NA,
    .default = 0))
#wlp %>% count(pboss)

## Management (excluding self-employers)
wlp$famgm <- case_match(wlp$wifamgm, c(1,2) ~ 1, c(97,99) ~ 9, .default = 0)
wlp$momgm <- case_match(wlp$wimomgm, c(1,2) ~ 1, c(97,99) ~ 9, .default = 0)
#xtabs(~ wifaself + famgm, wlp)

wlp <- wlp %>%
  mutate(famgm = case_when(
    wifaself == 2 | wifaself == 3 | wifaself == 4 | wifaself == 6 ~ 0,
    .default = famgm)) %>%
  mutate(momgm = case_when(
    wimoself == 2 | wimoself == 3 | wimoself == 4 | wimoself == 6 ~ 0,
    .default = momgm))
#xtabs(~ wifaself + famgm, wlp)
#xtabs(~ famgm + momgm, wlp)

wlp <- wlp %>%
  mutate(pmgm = case_when(
    famgm == 1 | momgm == 1 ~ 1,
    famgm == 9 & momgm == 9 ~ NA,
    .default = 0))
#wlp %>% count(pmgm)

wlp <- wlp %>% mutate(faboss = case_match(faboss, 9 ~ NA, .default = faboss)) %>%
  mutate(moboss = case_match(moboss, 9 ~ NA, .default = moboss)) %>%
  mutate(famgm = case_match(famgm, 9 ~ NA, .default = famgm)) %>%
  mutate(momgm = case_match(momgm, 9 ~ NA, .default = momgm))

## Educational Expectation (above college)
wlp$pexpedu <- case_match(wlp$wlp510a, c(4,5) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(pexpedu)

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## Book Exhibition
wlp$book <- case_match(wlp$wlp301, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(bookexpo)

## Art Performance
wlp$art <- case_match(wlp$wlp302, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(bookexpo)

## Volunteer
wlp$volunt <- case_match(wlp$wlp303, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(volunt)

## Relocation (actual or on paper)
wlp$reloc <- case_match(wlp$wlp501, 2 ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(reloc)

## Academic Communication
wlp$acadcomm <- case_match(wlp$wlp310, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(acadcomm)

## Talent Class
wlp$talentcls <- case_match(wlp$wlp304, 2 ~ 1, c(97,99) ~ NA, .default = 0)

wlp <- wlp %>% mutate(ntaxentcls =
  wlp3052 + wlp3053 + wlp3054 + wlp3055 + wlp3056) %>%
  mutate(ntaxentcls = case_match(ntaxentcls, 495 ~ 0, .default =ntaxentcls))
#wlp %>% count(ntaxentcls)

## Travel
wlp$travel <- case_match(wlp$wlp504, c(2,3) ~ 1, c(97,99) ~ NA, .default = 0)

## Abroad Study
wlp$abstudy <- case_match(wlp$wlp503, 2 ~ 1, c(97,99) ~ NA, .default = 0)

## Saving for Educaion
wlp$edusav <- case_match(wlp$wlp507, 2 ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(edusav)

## Select variables
wlp <- wlp %>% select(stud_id,
  marriage, faethn, moethn, pethn, # basic info.
  faedu, moedu, pedu, pedu2, income, income2, pses, # SES
  faocc, moocc, pocc, faboss, moboss, pboss, famgm, momgm, pmgm, # occupation
  pexpedu, # parenting
  book, art, volunt, reloc, # parenting-accompany

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acadcomm, talentcls, ntalentcls, travel, abstudy, edusav) # parenting-outsource

## Merge data
teps <- inner_join(wls, wlp, by = "stud_id") # n = 20055/19053 -> 19053

```

3. School Survey Data

```

## Choose not to use the school data, which has more missing values
#teps %>% distinct(wlsch_id) %>% count() # n = 333
#wlsch <- read_dta("data_raw/w1_j_sch_lv7.0.dta") # n = 329

## Attach school id to score data
score <- read_dta("data_raw/w1j_score.dta") %>% # n = 18929
  select(stud_id, score_fin) %>% drop_na() # n = 18032

score <- wls %>% select(stud_id, wlsch_id, wlppriv, wlurban3, wladmarea, geoarea) %>%
  right_join(score, by = "stud_id") # merge with wls to get school id

## Create a list of schools with their mean scores
schscore <- score %>% group_by(wlsch_id) %>% # create school list
  summarise(n = n(), meanscore = mean(score_fin), priv = first(wlppriv),
            urban = first(wlurban3), admarea = first(wladmarea), geoarea = first(geoarea)) %>%
  arrange(desc(meanscore)) %>% # sort by final score
  filter(n >= 10) # n = 332 -> 331

## Public and Private Schools by geoarea
teps %>% group_by(wlsch_id) %>%
  summarise(priv = first(wlppriv), admarea = first(wladmarea), geoarea = first(geoarea)) %>%
  xtabs(~ priv + geoarea, .)

##      geoarea
## priv  1  2  3  4  5  6
##      0 97 30 65 29 36 22
##      1 18  7  9 14  6  0

## Define Star Schools (top 54/279=19.35% of public schools)
schscore <- schscore %>% filter(priv == 0) %>%
  group_by(geoarea) %>%
  mutate(rank_bygeo = rank(-meanscore)) %>% # rank school by sector
  mutate(star = case_when(
    geoarea == 1 & rank_bygeo <= round(97*0.1935) ~ 1,
    geoarea == 2 & rank_bygeo <= round(30*0.1935) ~ 1,
    geoarea == 3 & rank_bygeo <= round(65*0.1935) ~ 1,
    geoarea == 4 & rank_bygeo <= round(29*0.1935) ~ 1,
    geoarea == 5 & rank_bygeo <= round(36*0.1935) ~ 1,
    geoarea == 6 & rank_bygeo <= round(22*0.1935) ~ 1,

```

```

    .default = 0)) %>%
  ungroup() %>%
  right_join(schscore)

```

```

## Joining with 'by = join_by(wlsch_id, n, meanscore, priv, urban, admarea,
## geoarea)'

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```

schscore %>% xtabs(~ star + geoarea, .)

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```

##      geoarea
## star  1  2  3  4  5  6
##      0 78 24 51 23 29 18
##      1 19  6 13  6  7  4

```

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schscore$star <- schscore$star %>% replace_na(0)

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## Merge data
teps <- schscore %>% select(wlsch_id, meanscore, star) %>%
  right_join(teps, by = "wlsch_id") %>% # n = 19053 -> 19053
  left_join(score) %>% # merge final scores
  arrange(stud_id)

```

```

## Joining with 'by = join_by(wlsch_id, stud_id, wlpriv, wlurban3, wladmarea,
## geoarea)'

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#teps %>% filter(is.na(star)) %>% distinct(wlsch_id) # two schools with final scores missing

teps$star <- tepts$star %>% replace_na(0)
#teps %>% filter(star == 1) %>% distinct(wlsch_id)

## Create Schooltype
teps$schtype <- case_when(
  tepts$wlpriv == 1 & tepts$star == 0 ~ 2,
  tepts$wlpriv == 0 & tepts$star == 1 ~ 3,
  .default = 1)
#teps %>% count(schtype)

```

4. Finalize Data

```

## Check missing data
#sapply(teps, function(y) sum(is.na(y))) %>% data.frame(na_count = .)

## Rename and order columns
teps <- tepts %>% rename(
  sch_id = wlsch_id,

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priv = w1priv,
urban = wlurban3,
scarea = w1scarea,
admarea = w1admarea,
finscore = score_fin)

teps <- teps %>% select(stud_id,
  sch_id, urban, scarea, admarea, geoarea, meanscore, priv, star, schtype, # school
  male, cogability, finscore, # student
  onlychild, edstchild, ygstchild, marriage, faethn, moethn, pethn, # family
  faedu, moedu, pedu, pedu2, income, income2, pses, # parental SES
  faocc, moocc, pocc, faboss, moboss, pboss, famgm, momgm, pmgm, # parental occupation
  pexpedu, # parenting
  book, art, volunt, reloc, # parenting-accompany
  acadcomm, talentcls, ntalentcls, travel, abstudy, edusav) # parenting-outsource

## Export data
write_dta(teps, "data/TEPS3.dta")

```
